

INVITED TALK / ABSTRACT

Higher Codimensional Isoperimetric Problems

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Abstract

Motivated by Almgren's higher-codimensional isoperimetric problems, we investigate the existence of minimal surfaces in Euclidean three-space whose boundary has constant geodesic curvature and vanishing normal curvature. I will present several existence results and discuss the parallel with the well developed theory of constant mean curvature surfaces. This is ongoing joint work with L. Hauswirth, T. Thalmaier, and M. Mamadou Idrissou.

Keywords: higher codimensional isoperimetric problems; minimal surfaces; Euclidean three-space; constant geodesic curvature; vanishing normal curvature; existence results; constant mean curvature surfaces